



General instructions

1. Read all the questions and understand them clearly before you start writing the programs. In case of any doubts, get them clarified with your invigilator.
2. The input should not be hard coded inside the program. The program should work with any input without changing the code for each input.
3. Programs logic, coding standards, modularity will be considered for evaluation.
4. Programs should work with all the sample inputs given in this sheet.
5. Any language can be used for writing the programs.
6. You are expected to consider all the edge cases when solving the problem.



Problem 1

You are given an array that has only zeros and ones. Your task is to find the difference between the **maximum and the minimum number of consecutive ones** in the array.

The first line contains a single integer N . The next line has N integers.

Sample Input

12

0 1 1 0 0 1 0 1 1 1 1 0

Sample Output

3

Problem 2

Given an array of integers, all numbers occur exactly twice except one number which occurs once. Find the number in **$O(N)$ time**.

Sample Input

7 3 5 4 5 3 4

Sample Output

7

Note: Added benefit if no extra space is occupied.



Problem 3

Find whether the given mathematical expression has balanced parentheses "()", braces "{}" and square brackets "[]".

Sample Input 1:

(2+{3})

Sample Output 1:

YES

Sample Input 2:

{{2+3}

Sample Output 2:

NO

(42+3)}

Problem 4

Given a string S, determine whether any permutation of it is a palindrome.

Sample Input 1:

radar

Sample Output 1:

YES

Sample Input 2:

Zoho

Sample Output 2:

NO

}

({ { { }



Sample Input 3:

daarr

Sample Output 3:

YES

Problem 5

Find if a string A can be formed by removing a few characters from the other given string B. If so String A becomes a subsequence of String B. A is said to be a subsequence of B, if A can be obtained by deleting some characters of B without changing the order of the remaining characters in B. Determine if A is a subsequence of B.

Sample Input 1:

Set Sangeet

Sample Output 2:

YES

Sample Input 2:

Zoho India

Sample Output 2 :

NO

Sample Input 1:

Set Step

Sample Output 2 :

NO

Problem 6

You have three stacks of cylinders where each cylinder has the same diameter, but they may vary in height. You can change the height of a stack by removing and discarding its topmost cylinder any number of times.

Find the maximum possible height of the stacks such that all of the stacks are exactly the same height. This means you must remove zero or more cylinders from the top of zero or more of the three stacks until they are all the same height, then print the height. The removals must be performed in such a way as to maximize the height.

Note: An empty stack is still a stack.

Input Format

The first line contains three space-separated integers n_1 , n_2 , n_3 describing the respective number of cylinders in stacks 1, 2, 3. The subsequent lines describe the respective heights of each cylinder in a stack from top to bottom:

The second line contains n_1 space-separated integers describing the cylinder heights in stack .

The third line contains n_2 space-separated integers describing the cylinder heights in stack .

The fourth line contains n_3 space-separated integers describing the cylinder heights in stack .

Sample Input 1:

5 3 4

3 2 1 1 1

4 3 2

1 1 4 1

Sample Output 1:

5